MIL-C-81706
Amendment 4
10 March 1977
Superseding
Amendment 3
31 August 1972

MILITARY SPECIFICATION

CHEMICAL CONVERSION MATERIALS FOR COATING
ALUMINUM AND ALUMINUM ALLOYS

This amendment forms a part of Military Specification MIL-C-81706, dated 30 June 1970, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1

*1.2.2.1 Add:

"Form IV: Premixed liquid, thixotropic (ready for use)."

"Form V: Premeasured powder, thixotropic (ready for use after addition of water)."

Page 2

2.1 Delete: "L-B-560 Bottle, Screw Cap (Polyethylene)" and substitute: "MIL-P-26701, Bottle, Screw Cap and Carboys, Polyethylene Plastic".

Delete: "TT-P-143" and substitute: "PPP-P-1892".

Page 4

*3.4 Add:

"Type IV and Type V films shall show an average weight of 2.8 to 5.5 of material adhering to panels tested in accordance with the method specified in 4.5.7. The material shall cling uniformly with no visual evidence of excessive run-off or lack of film coverage."

Page 5

*3.10 Delete from Form I and Form II: "6 months" and substitute: "12 months".

Add:

"Form IV - Premixed liquid, thixotropic. 12 months"

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Page 5, 3.10 (cont)

"Form V - Premeasured powder, thixotropic. 12 months"

Page 6

4.2 Add: "(c) Periodic qualification reevaluation (see 4.8)".

*4.3.2 Delete: "a minimum of five gallons of solution" and substitute: "a minimum of eight one quart bottles of solution".

Page 7

*4.3.3.1 Add new paragraph 4.3.3.1.1:

"4.3.3.1.1 Panel Preparation for Adherence Test. Three 3 inches x 10 inches x .032 inches panels of 7075-T6 aluminum alloy shall be cleaned with benzene followed by acetone. (3 inch x .032 inches x any length over 4 inches may be used.) The panels shall then be marked with a pencil line 4 inches from one end. Each clean, marked panel shall be placed in a separate 600 ml beaker and weighed with the beaker to the nearest tenth of a gram."

Page 8

Add to title of Table II: "AND QUALIFICATION REEVALUATION".

Page 13

*4.5 Add the following new paragraphs:

"4.5.7 Vertical Adherence Test. In addition to the other tests required for Class 1A materials, Form IV and V materials shall be tested for ability to adhere to a vertical surface as specified in 3.4."

"4.5.7.1 Material Preparation. One quart of the Form IV or Form V material (with the required water added) shall be thoroughly mixed and allowed to stand for 24 hours. Enough of the material to almost fill the beaker shall then be transferred to a 600 ml plastic beaker."

"4.5.7.2 Adherence. Each previously prepared panel, in accordance with 4.3.3.1.1, shall be immersed, one at a time, in the coating material up to the 4" pencil line, lifted vertically above the

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Page 13, 4.5 (4.5.7.2) (cont)

surface of the material and allowed to drain for 45 seconds. The panel shall then be transferred without rinsing to the beaker in which it was originally weighed and the beaker and coated panel shall be weighed again to the nearest tenth of a gram."

"4.5.7.3 Weight of material clinging. The weight of material clinging to the panel is equal to the original weight of the panel plus the beaker subtracted from the weight of the panel plus the coating plus the beaker. Average weight determined from the three tests shall conform to 3.4."

Page 17

Add the following new paragraphs:

"4.8 Periodic qualification reevaluation. This reevaluation shall be made at two-year intervals after date of letter of notification of the material's acceptability for qualification. Materials from current production shall be evaluated against the requirements listed in Table II in addition to the quality conformance requirements listed in Table III."

"4.8.1 Reevaluation procedure. It shall be the responsibility of the supplier of a qualified product to furnish to the Government, at two-year intervals, the data necessary to establish the continued conformity of his material to all qualification requirements. These data shall preferably be complete test results of a sample representing current production, tested against all the requirements of the specification. At the discretion of the qualifying activity, test records from current production may be accepted for the reevaluation to the extent they are available, and samples from current production need be subjected to only the tests for which no production test records are available. The qualifying activity shall be notified of the test results. If a failure occurs, no chemical conversion materials for coating aluminum and aluminum alloys represented by the sample, nor any other material produced with the same ingredients and processes, which has not been submitted for quality conformance inspection, shall be offered for acceptance until the cause of failure has been determined and concurred in by the qualifying activity as not affecting the ability of the coating material to pass qualification inspection requirements. In the event the date for reevaluation has passed and no current production materials or data are available for reevaluation, the supplier will still be eligible for contract award, but acceptance of material from such a supplier will be provisional and will be final only after a sample of his material has met all the qualifying requirements of the specification."

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Page 17 (cont)

5.2 Delete: "Specification TT-P-143" and substitute" PPP-P-1892".

5.2.1 Delete fourth sentence and substitute: "Polyethylene bottles with polyethylene lined screw cap closures, conforming to MIL-B-26701, shall be used for packaging of premixed liquids for touch-up brush application and for powder in quantities of 5 pounds or less."

* Delete: "(see 6.2)" and substitute" "and the premeasured powder shall be packaged in one quart quantities with the bottle marked with the powder level and the added liquid level (see 6.2)."

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6.7 Add new paragraph 6.7.1:

"6.7.1 Conformity to qualified sample. It is understood that chemical conversion materials supplied under contract shall be identical in every respect to the qualification sample tested and found satisfactory, except for changes, if any, previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection."

6.8 Changes from previous issue. The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

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